

Class XII | Unit 7 | weight 10%

Wireless and mobile communication

Board weight about 10% | 15 theory periods

Wireless communication carries data without a cable between the devices. Mobile communication is wireless communication that continues while the user moves. This lecture compares the short-range technologies, the satellite orbits, and the cellular network that a phone actually uses.

What you should be able to do

- Define wireless and mobile communication and the basic radio terms.
- Compare Wi-Fi, WiMAX, Bluetooth, infrared and microwave on range, speed and frequency.
- Describe long-distance wireless links and the GEO, MEO and LEO orbits.
- Explain cells, base stations, frequency reuse, MSC and BTS.
- Describe the generations of mobile networks and distinguish a fixed ad hoc network from a MANET.

Path through the lecture

- 1 Separate wireless from mobile with two examples: a fixed microwave link, and a phone in a moving bus.
- 2 Put the short-range table on one board. It is a favourite short question.
- 3 Draw three hexagons, a BTS in each, and one MSC. Then attach the generations to that drawing.

7.1 Terms

Wireless communication. Communication that uses electromagnetic waves rather than a cable between the sender and the receiver.

Mobile communication. Wireless communication that allows the user to move and to keep the connection, typically through a cellular network.

- The **frequency spectrum** is the range of frequencies available for radio signals. Different services are given different bands so they do not all talk on top of each other.
- A **radio signal** is the electromagnetic wave that carries the data.
- A **radio transceiver** both transmits and receives. A phone contains one. The name joins transmitter and receiver.
- An **access point** is the device that joins wireless users to a wired network. A home or lab Wi-Fi box is an access point, often combined with a router.
- **Line of sight** means the path between two antennas is clear. Microwave links need it. A hill or a building in the path interrupts them.

7.2 Short-range technologies

The figures below are the textbook comparisons used for this unit. A particular product can differ. In an answer, the order matters more than a single brand's specification: Bluetooth is personal, Wi-Fi is a room or a building, WiMAX was designed for a city, infrared is a short beam, and microwave can jump between distant towers when the path is clear.

Short-distance wireless technologies

Technology	Typical range	Speed in this comparison	Frequency band
Bluetooth	About 10 metres, more for some classes	Lower than Wi-Fi, enough for headsets and small file transfers	2.4 GHz
Infrared	About a metre to a few metres, and only in a clear beam	Low, suitable for a remote control or a short link	Infrared light, above radio frequencies
Wi-Fi	Tens of metres indoors, more in the open	High, from many megabits per second upward	2.4 GHz and 5 GHz
Microwave	Kilometres between aligned antennas	High on a point-to-point link	Microwave bands, and the path must be line of sight
WiMAX	A metropolitan area, up to tens of kilometres in the design	Broadband, lower or comparable to later Wi-Fi in practice	Licensed bands around 2 to 11 GHz in the original designs

7.3 Long distance and satellites

Long-distance wireless communication crosses cities, countries or oceans by radio towers, microwave chains or satellites. A satellite receives a signal on one frequency, the uplink, and sends it back toward Earth on another, the downlink. The height of the orbit decides the delay, the coverage and the number of satellites needed.

Orbits used with satellite communication and navigation

Orbit	Height and behaviour	What to remember
GEO, geostationary	About 36,000 km above the equator. The orbital period matches the Earth's rotation, so the satellite stays over the same region.	One satellite covers a large fixed area. The signal has a noticeable delay. Weather and television links use this idea.
MEO, medium Earth orbit	Roughly 2,000 to 20,000 km.	Navigation constellations such as GPS use MEO, around 20,000 km. Coverage of the globe needs a group of satellites, fewer than a low orbit needs.
LEO, low Earth orbit	Below about 2,000 km.	The delay is smaller and the satellite moves quickly across the sky. Continuous coverage needs many satellites. The curriculum's phrase 'types of GPS' is answered with these three orbits, and you should add that the GPS constellation itself flies in MEO.

7.4 The cellular network

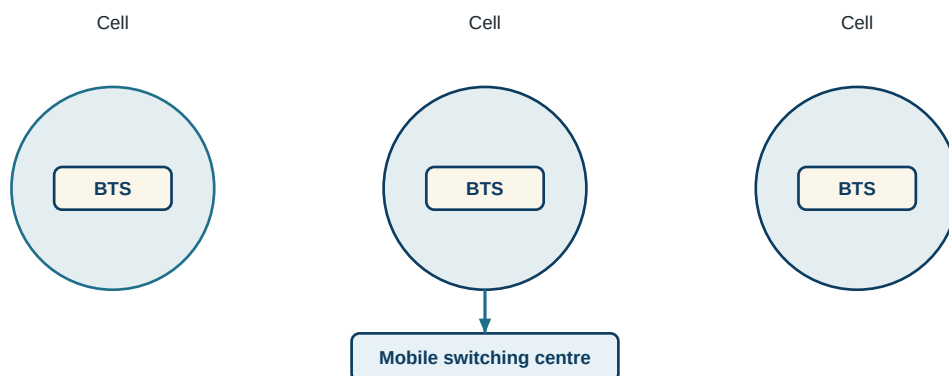


Figure 7.1 Cells, a base transceiver in each, and a mobile switching centre.

- A **cell** is the area served by one base station. Diagrams draw it as a hexagon so the cells tile a map.
- A **base transceiver station (BTS)** is the radio equipment and antenna in that cell. It is the part the phone talks to.

- A **base station** is the broader name for the equipment that serves a cell, including the BTS and its controller.
- **Frequency reuse** means the same channel frequencies can be used again in cells that are far enough apart. Neighbouring cells use different frequencies so their calls do not collide. That reuse is why a limited spectrum can serve a whole city.
- The **mobile switching centre (MSC)** connects cells to each other and to the ordinary telephone network. It tracks which cell a phone is in and supports the handover when a caller crosses a boundary.
- The **interface between cells** is the handover: the call is passed from one base station to the next while the conversation continues.

7.5 Generations

Each generation is a family of technologies for the cellular network. The 2019 curriculum pairs the names as 2G (GSM), 3G (GPRS), 4G (LTE) and 5G. Write those pairs if the question uses them. For a precise answer, add the engineering names: GSM is 2G voice and digital service, GPRS is the packet-data step often called 2.5G, 3G is the UMTS family, 4G is LTE, and 5G is the following generation. The curriculum's bracket '5G (Future)' reflects the year of the curriculum. 5G networks have since been deployed in many countries. What you should say about 5G is that it is designed for higher data rates and lower delay than LTE, and for a much larger number of connected devices.

Generations

Name in the question	What the user gained
2G GSM	Digital mobile calls and text messages.
GPRS, listed with 3G in the curriculum note	Packet data over the mobile network, so the phone could carry internet traffic. Technically this step is 2.5G, and 3G then raised the data rate further.
4G LTE	Much higher data rates for video and ordinary internet use on the phone.
5G	A further rise in rate and a fall in delay, and support for many more devices.

7.6 Ad hoc networks

Ad hoc network. A network formed by devices that communicate directly and cooperate to forward data, without depending on fixed infrastructure such as an access point or a base station.

A **fixed ad hoc network** is set up for a time among nodes that do not move, for example laptops in a hall that forward for one another. A **mobile ad hoc network (MANET)** is an ad hoc network whose nodes move. Routes change as devices come and go, so the network has to rediscover paths. A disaster team whose radios form a network without a tower is the example to give.

Exam point

Keep three comparisons ready: Wi-Fi with Bluetooth, GEO with LEO, and a cellular network with a MANET. Each is a 3-mark table.

Key terms

Term	Meaning to write in an answer
Wireless	Communication by electromagnetic waves, without a data cable between the ends.
Access point	A device that connects wireless stations to a network.

Term	Meaning to write in an answer
Line of sight	A clear path between antennas.
Cell	The geographic area served by one base station.
BTS	The radio transmitter and receiver that serves a cell.
MSC	The switch that connects cells and handles handover.
GEO	A geostationary orbit, fixed relative to the ground.
MANET	A mobile ad hoc network, with moving nodes and no fixed infrastructure.

Bring it together

- Wireless removes the cable. Mobile communication also survives movement.
- Bluetooth, infrared, Wi-Fi, microwave and WiMAX differ in range, speed and band.
- GEO stays over one region at great height. GPS uses MEO. LEO needs many fast-moving satellites.
- A cellular network reuses frequencies across cells and hands a call over at the boundary through the MSC.
- GSM, GPRS, LTE and 5G are the generation story. An ad hoc network has no fixed base station. A MANET also has moving nodes.

Do this in the practical

- 1 From the lab Wi-Fi, identify the access point name and compare its range with a Bluetooth headset in the same room.
- 2 Draw a three-cell sketch of the college area and mark where a handover would happen if a caller walked between them.

Check yourself

Attempt every item before you read the key. Section A of the Computer Science paper is a set of multiple-choice questions.

1. A transceiver:

- A. Transmits and receives
- B. Only prints
- C. Stores a primary key
- D. Compiles C++

2. Infrared communication typically requires:

- A. A short, clear beam
- B. A geostationary satellite
- C. A coaxial backbone across a city
- D. An MSC

3. Wi-Fi is described by the family:

- A. IEEE 802.11
- B. ASCII
- C. 1NF
- D. BIOS

4. A geostationary satellite:

- A. Stays over the same region of the Earth
- B. Orbits inside the classroom
- C. Is a type of RAM
- D. Replaces the SIM card

5. GPS satellites operate in:

- A. Medium Earth orbit
- B. The CMOS battery
- C. A register of the CPU
- D. A text file

6. Frequency reuse in cellular networks allows:

- A. The same frequencies to be used again in distant cells
- B. One frequency for the entire world with no planning
- C. Removal of the need for a phone
- D. Text files to become binary

7. The MSC is responsible for:

- A. Switching calls and supporting handover between cells
- B. Erasing EPROM
- C. Compiling a class
- D. Drawing a flowchart oval

8. A MANET differs from a fixed ad hoc network because:

- A. Its nodes move
- B. It uses only fibre
- C. It is a normal form
- D. It cannot forward data

Answer key

Q	Answer	Why this is the answer
1	A	The device combines a transmitter and a receiver.
2	A	Infrared is short range and does not pass through walls.
3	A	IEEE 802.11 is the wireless LAN standard behind Wi-Fi.
4	A	Its period matches the Earth's rotation.
5	A	The GPS constellation uses MEO.
6	A	Cells far apart can reuse channels, which multiplies capacity.
7	A	The mobile switching centre connects the cells and the wider telephone network.
8	A	MANET means a mobile ad hoc network.

Short answers

A 3-mark answer needs a definition and two clear points, or a three-row difference. Write in sentences.

Q1. Differentiate wireless communication and mobile communication.

Model answer

Wireless communication sends data by electromagnetic waves without a cable between the ends. A fixed microwave link is wireless and the ends stay still. Mobile communication is wireless communication that continues while the user moves, as in a cellular phone call. Mobile communication is therefore a moving form of wireless communication.

Q2. Compare Wi-Fi and Bluetooth on range and use.

Model answer

Bluetooth is a short personal link, about 10 metres, used for a headset or a small accessory, in the 2.4 GHz band. Wi-Fi covers a room or a building, carries much more data, and joins devices to a local network through an access point, using 2.4 GHz or 5 GHz. Wi-Fi is the network. Bluetooth is the short personal connection.

Q3. Differentiate GEO and LEO.

Model answer

A GEO satellite orbits at about 36,000 km and stays over one region, so a fixed dish can use it, at the cost of a longer delay. A LEO satellite orbits below about 2,000 km, moves quickly across the sky and has a shorter delay. Continuous service from LEO needs a large constellation.

Q4. What is a cell, and why do neighbouring cells use different frequencies?

Model answer

A cell is the area covered by one base station. Neighbouring cells use different frequencies so that calls near the boundary do not interfere. The same frequency can be reused in a cell that is farther away. That is frequency reuse.

Q5. Define an ad hoc network and a MANET.

Model answer

An ad hoc network is formed by devices that talk directly and forward data for one another, without a fixed access point or base station. A MANET is an ad hoc network whose nodes move, so the routes have to be found again as the topology changes.

Detailed answers

Section C asks for a connected answer. Open with a definition, develop the points below, and close with one example.

Question 1. Explain the parts of a cellular network and the idea of frequency reuse and handover.

- 1 Define wireless and mobile communication.
- 2 Define cell, BTS, base station and MSC, and describe how a phone reaches the wider network.
- 3 Explain frequency reuse with neighbouring cells on different channels.
- 4 Explain handover as the caller crosses a cell boundary.
- 5 Add one sentence on 2G, 4G and 5G so the network has a technology as well as a map.

Question 2. Compare short-distance wireless technologies and the three satellite orbits.

- 1 Give the comparison table for Bluetooth, infrared, Wi-Fi, microwave and WiMAX: range, speed and frequency.
- 2 Define line of sight and attach it to microwave and infrared.
- 3 Describe GEO, MEO and LEO with height and one use.
- 4 State that GPS flies in MEO.

- 5 Close with the access point as the device that joins a short-range Wi-Fi user to the wired network.